

Finding derivatives — 6.4

Example: Find the 100^{th} derivative of $x^2 \cos(x)$ evaluated at $x=0$.

Instead of directly differentiating, we can use a Taylor series!

$$x^2 \cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n+2}$$

$= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$ the 100^{th} derivative at $x=0$ is almost the coefficient of x^{100} , we're just off by 100!

We want the coefficient of x^{100} .

$$x^2 \cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n+2}$$

We want c_{100} ,
and to get the
 $c_{100} x^{100}$ term, we

need $2n+2 = 100$, thus $2n = 98$, so $n = 49$

$$c_{100} x^{100} = \frac{(-1)^{49}}{98!} x^{100}, \text{ and therefore}$$

$$\frac{f^{(100)}(0)}{100!} = c_{100} = \frac{(-1)^{49}}{98!}$$

$$\Rightarrow f^{(100)}(0) = 100! \cdot \frac{-1}{98!} = -1 \cdot 100 \cdot 99 = -9900.$$

In the same way,

$$f^{(101)}(0) = 101! c_{101} = 0 \quad (\text{series only has even powers of } x)$$

$$f^{(102)}(0) = 102! c_{102} = 102! \frac{(-1)^{50}}{100!} = 102 \cdot 101$$